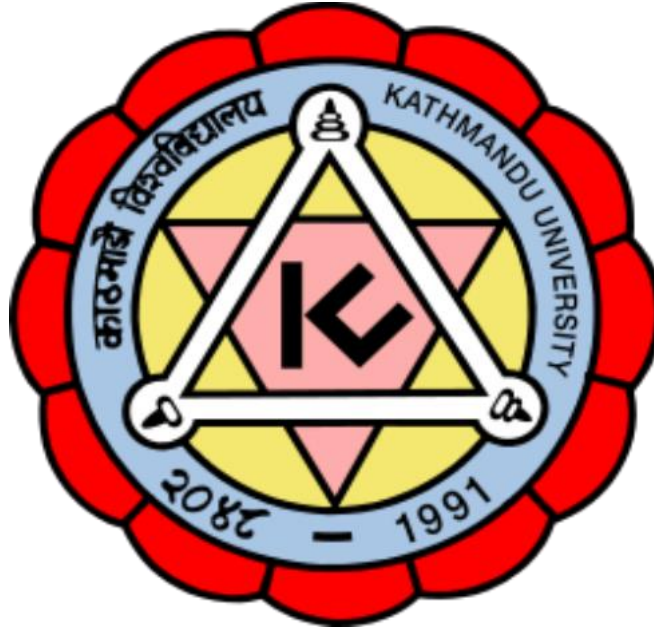


KATHMANDU UNIVERSITY SCHOOL OF ENGINEERING

DEPARTMENT OF HEALTH INFORMATICS

DHULIKHEL, KAVRE



A CONCEPT NOTE ON

Maternal and Child Health Tracking System

(CODE NO: HIMS256)

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1. Background

Maternal and child health remains a major global public health concern, especially in low- and middle-income countries (LMICs). Despite significant progress, preventable maternal and neonatal deaths continue due to delayed antenatal care (ANC), inadequate postnatal follow-up, poor data tracking, and weak health information systems.

Globally, maternal mortality ratio (MMR) and under-five mortality remain high in several regions, including South Asia. A major contributing factor is fragmented health data systems that fail to ensure continuity of care across pregnancy, delivery, and postnatal stages.

Studies show that improving health information systems and digital tracking tools can significantly enhance ANC attendance, skilled birth delivery, immunization coverage, and early detection of complications.

In Nepal, maternal death surveillance systems exist, but challenges remain in real-time monitoring, data integration between community and health facilities, and timely decision-making. This highlights the need for an integrated digital Maternal and Child Health Tracking System.

Therefore, a comprehensive MCH Tracking System is needed to strengthen monitoring, improve service delivery, and reduce maternal and child mortality.

2. Problem Statement

Many healthcare facilities still use paper-based systems for maternal and child health records. These systems create several challenges, including:

- Loss or damage of patient records
- Difficulty in tracking pregnant women and children
- Missed vaccination schedules and appointments
- Delayed reporting and decision-making
- Lack of proper communication between health workers and patients
- Inaccurate data management and duplication

Because of these problems, maternal and child mortality rates may increase, and healthcare services become less efficient. Therefore, there is a need for a computerized tracking system that can improve record management, monitoring, and healthcare delivery .

3.Objectives

3.1General objective

To develop a digital Maternal and Child Health Tracking System for improving healthcare service delivery, monitoring, and follow-up of pregnant women and children.

3.2Specific Objectives

1. To register and maintain records of pregnant women
2. To track Ante-Natal Care (ANC) services
3. To manage delivery and Post-Natal Care (PNC) services
4. To register children and monitor immunization schedules
5. To generate reminders and notifications through SMS

4.Literature review

Maternal and child health is an important public health issue, especially in developing countries. Many healthcare facilities still use paper-based systems, which can lead to loss of records, poor data management, delayed reporting, and difficulty in tracking pregnant women and children.

Studies show that digital health information systems improve healthcare services by maintaining accurate records, tracking ANC and immunization schedules, and sending reminders to patients. Mobile technologies such as SMS notifications also help improve communication between healthcare workers and patients.

Countries like India have implemented digital Maternal and Child Tracking Systems to improve maternal and child healthcare services. In Nepal, challenges still exist in monitoring and managing maternal and child health data effectively.

Therefore, a digital Maternal and Child Health Tracking System is needed to improve record management, monitoring, follow-up services, and healthcare delivery.

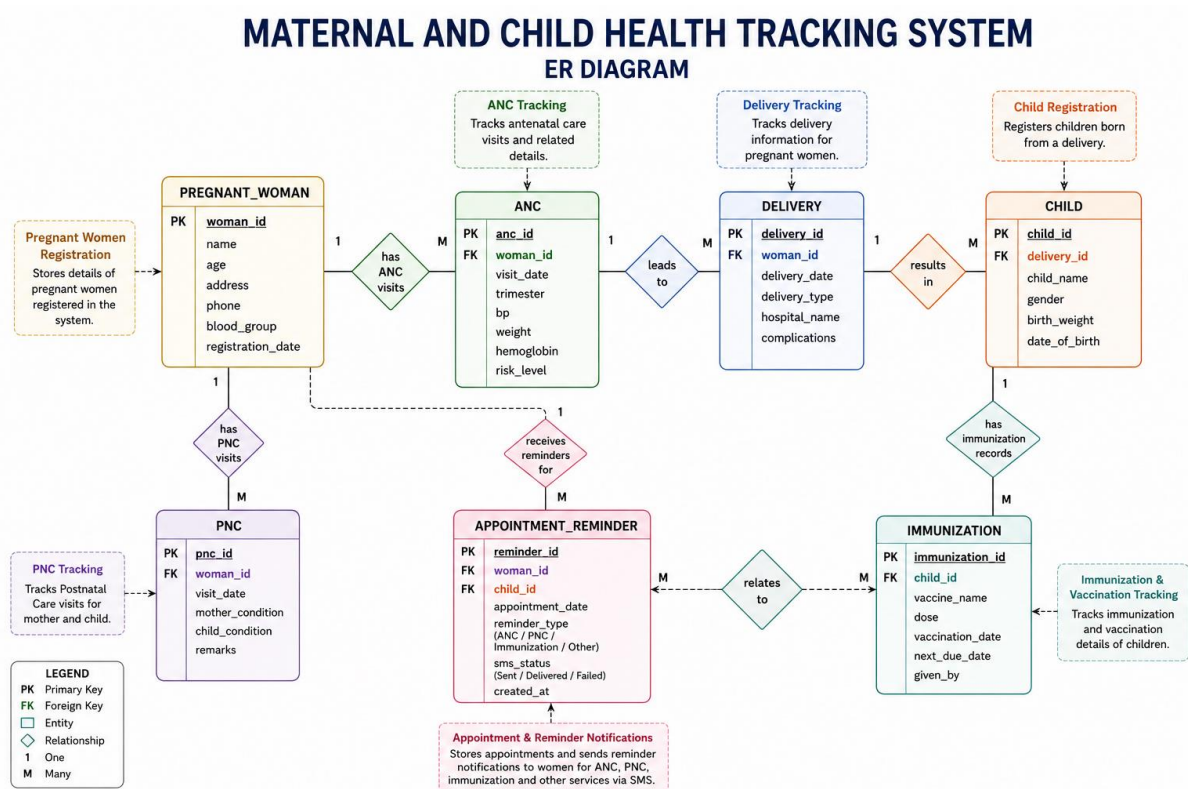
5.System features

- Pregnant women registration
- Child registration
- ANC, delivery, and PNC tracking
- Immunization and vaccination tracking
- Appointment and reminder notifications through SMS

6. Methodology

1. **Requirement Analysis:** Collect information and identify system requirements from healthcare workers and users.
2. **System Design:** the database, user interface, and system workflow.
3. **System Development:** Develop the Maternal and Child Health Tracking System using suitable technologies.
4. **Testing:** Test the system for errors, functionality, and security.
5. **Implementation:** Deploy the system and provide training to healthcare staff.
6. **Maintenance and Evaluation:** Monitor system performance and update the system when required.

7. System design



8.Tools and Technologies

The system can be developed using:

Component	Technology
Frontend	HTML, CSS, Bootstrap
Backend	Django, python
Database	MySQL
Platform	Web based system
Additional tools	SMS notification system

9.Conclusion

Maternal and child health is a key area of public health that requires proper monitoring and timely healthcare services. The proposed Maternal and Child Health Tracking System aims to improve the management of maternal and child health information by digitizing records and tracking essential services such as ANC, delivery, PNC, and immunization. This system will help reduce missed appointments, improve data accuracy, and support healthcare workers in providing better services. Overall, it will contribute to improving maternal and child health outcomes in Nepal.

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